Educational Data Mining

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CSE 408

WHAT IS DATA MINING

Data Mining is the process of discovering **hidden patterns**, **correlations**, **trend** and **useful information** from large dataset using machine learning, statistics and database systems [1]

- Converts **raw** and **unstructured** data into meaningful and **actionable information**
- About ~**2.5 quintillion bytes** generated daily [2]
- Organizations are “data rich but information poor”
- Bridge the gap by turning data into insights that **support decision-making**
- **1 Zettabyte = 1 Trillion Gigabytes**

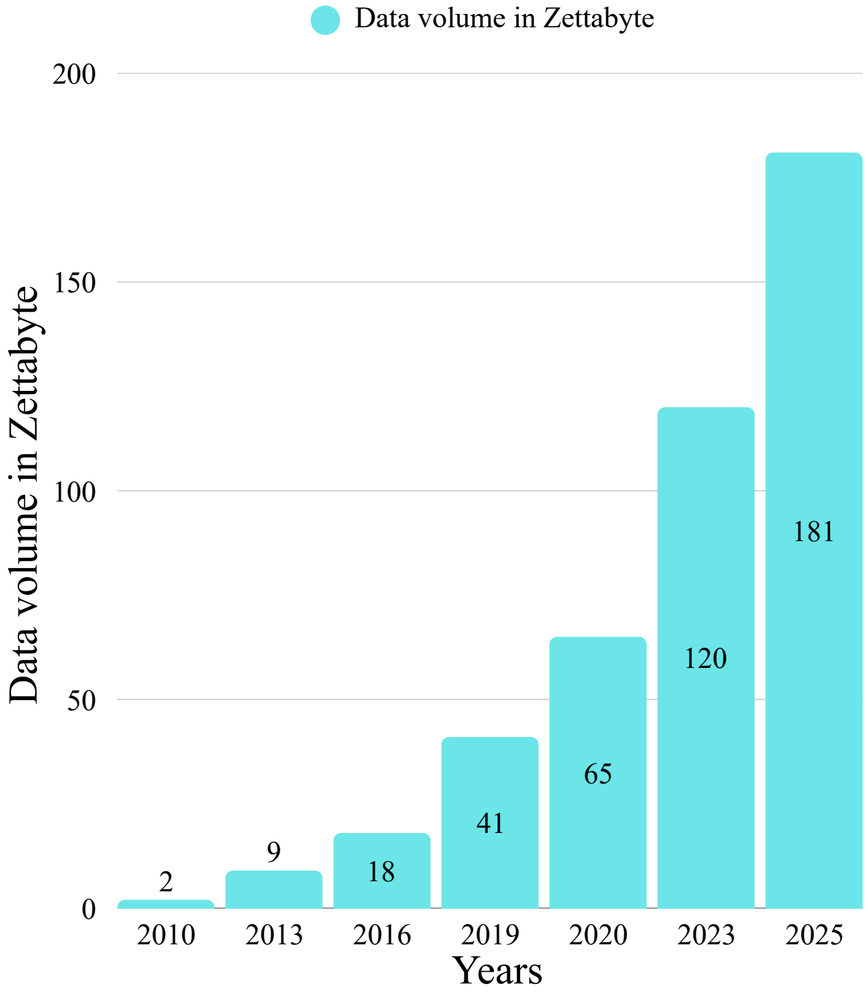


Fig 1:Volume of data or information created, captured, copied, and consumed worldwide from 2010 to 2025 [2]

WHAT IS EDUCATIONAL DATA MINING (EDM)

Educational Data Mining is the process of extracting meaningful patterns and knowledge from educational data to better understand learning processes and enhance educational decision making.[3]

- Applies data mining techniques to **educational data**
- Analyzes students' **learning behavior** and **academic performance**
- Supports data-driven **educational decision making**
- Identifies **at-risk students** and learning difficulties early
- The ultimate goal of EDM is to improve **teaching strategies**, **learning outcomes**, and **educational quality**

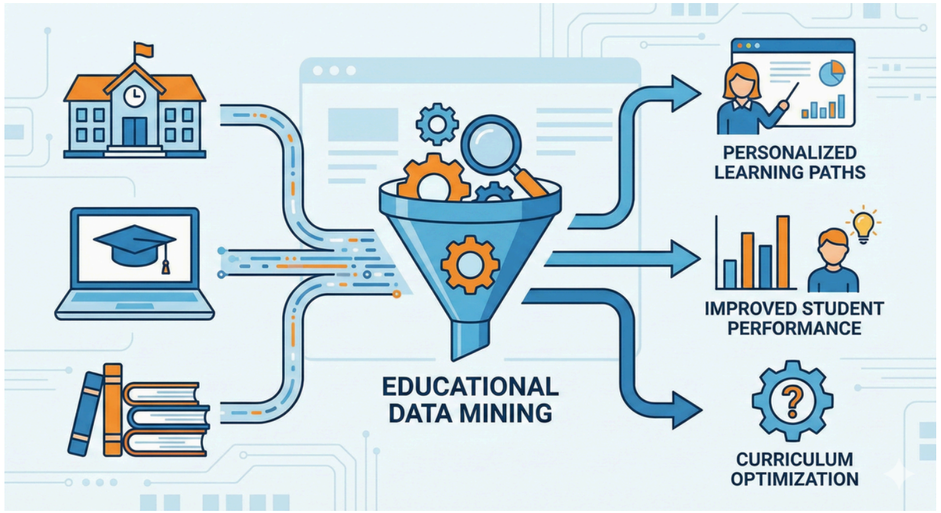


Fig 2: EDM image

EDM PROCESS

KDD (Knowledge Discovery in Databases) Process [4]

- **Data Selection** – Selecting relevant data required for analysis
- **Data Integration** – Combining data from multiple sources into a unified dataset
- **Data Cleaning** – Removing noise and missing values
- **Data Transformation** – Converting data into appropriate formats for mining
- **Data Mining** – Applying algorithms to extract patterns and relationships
- **Pattern Evaluation** – Identifying meaningful and useful patterns
- **Knowledge Representation** – Presenting discovered knowledge in an understandable form

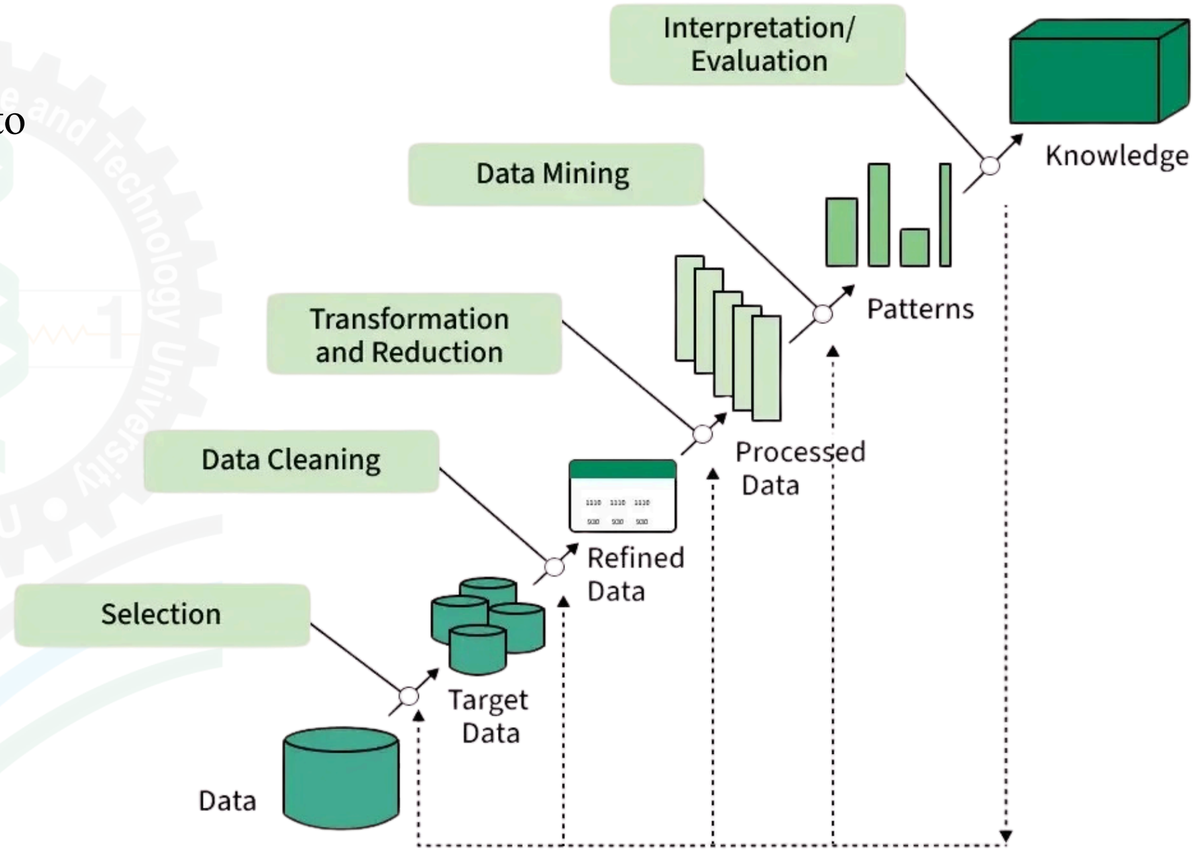
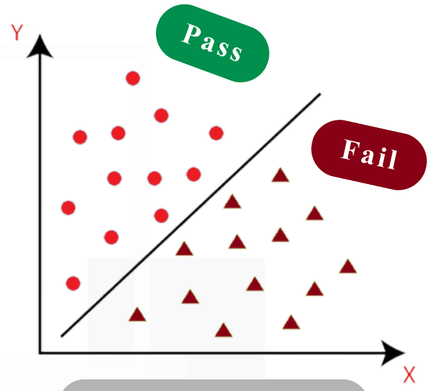


Fig 3: Knowledge Discovery in Databases

COMMON TECHNIQUES OF EDM [5]

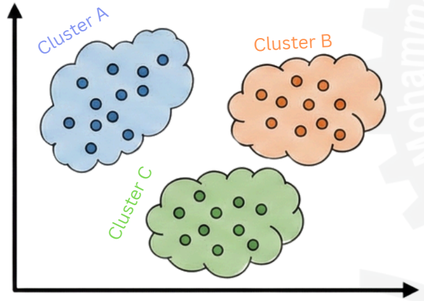
Classification



CLASSIFICATION

Example: Predicting whether a student will Pass or Fail based on quiz scores.

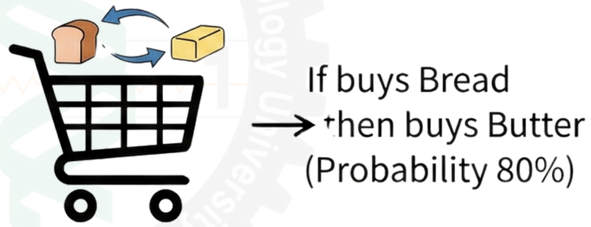
Clustering



CLUSTERING

Example: Segmenting students into high, medium, and low performers.

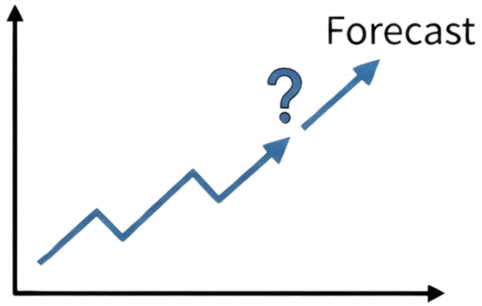
Association Rules



ASSOCIATION RULES

Example: Students who study late at night often submit assignments late.

Prediction

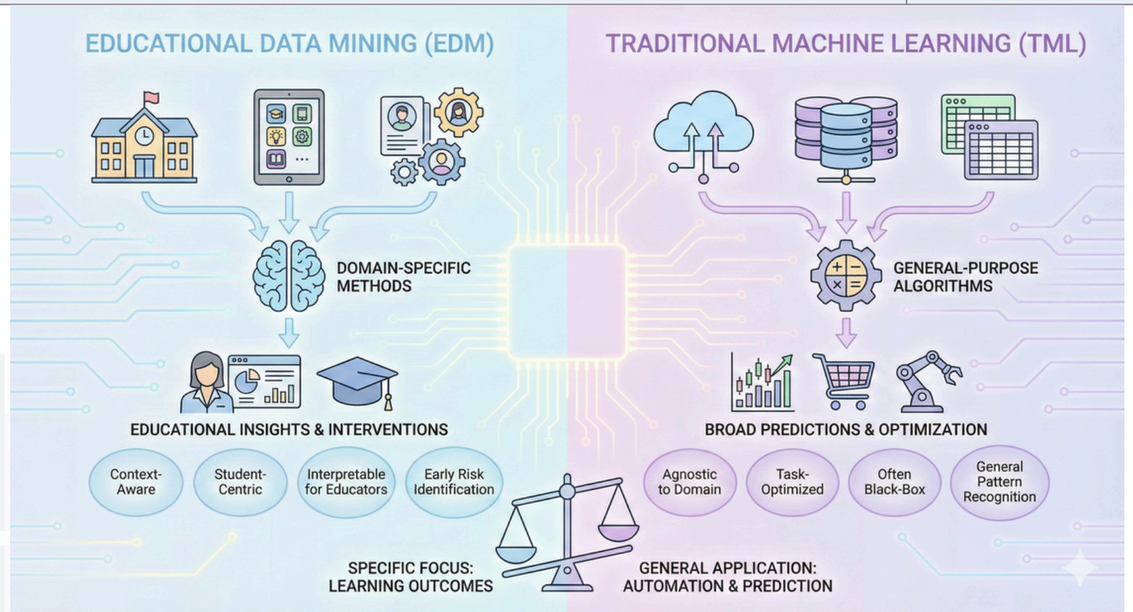


PREDICTION

Example: Predicting a student's final grade based on past performance.

Comparison table EDM vs Traditional ML [6]

Aspect	Educational Data Mining (EDM)	Machine Learning (ML)Method
Main Purpose	Improve student learning and academic performance	Enable systems to learn from data and make predictions
Primary Focus	Understanding and improving learning and education	Model training, Prediction, classification, and automation
Type of Data Used	Student grades, attendance, LMS activity, assignments	Any structured or unstructured data
Goal	Improve teaching strategies and learning outcomes	Improve model accuracy and performance
Outcome	Early identification of at-risk students	Accurate predictions or decisions
Users	Educators, researchers, institutions	Data scientists, engineers
Interpretability	High – results are easy for students and teachers to understand	Often low – complex or black-box models
Example	Identifying at-risk students	Image recognition, spam detection



Key Importance of EDM



Handle Large Educational Data



Understand learning partterns



EDM



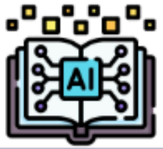
Bridge the Information gap



Improve Decision Making



Support early Intervention



Adapt to personalized learining needs

Real life applications of EDM

1. Student Performance Analysis

Final grade prediction accuracy reaches 88.6% using current academic records [7].

2. Early Warning Systems

EDM-based early warning systems reduce dropout rates by 14% [8].

3. Personalized & Adaptive Learning

Adapting learning materials based on individual student needs and behavior.

4. Curriculum & Teaching Improvement

Evaluating teaching methods and course design through analysis of LMS data.

5. Institutional Decision Making

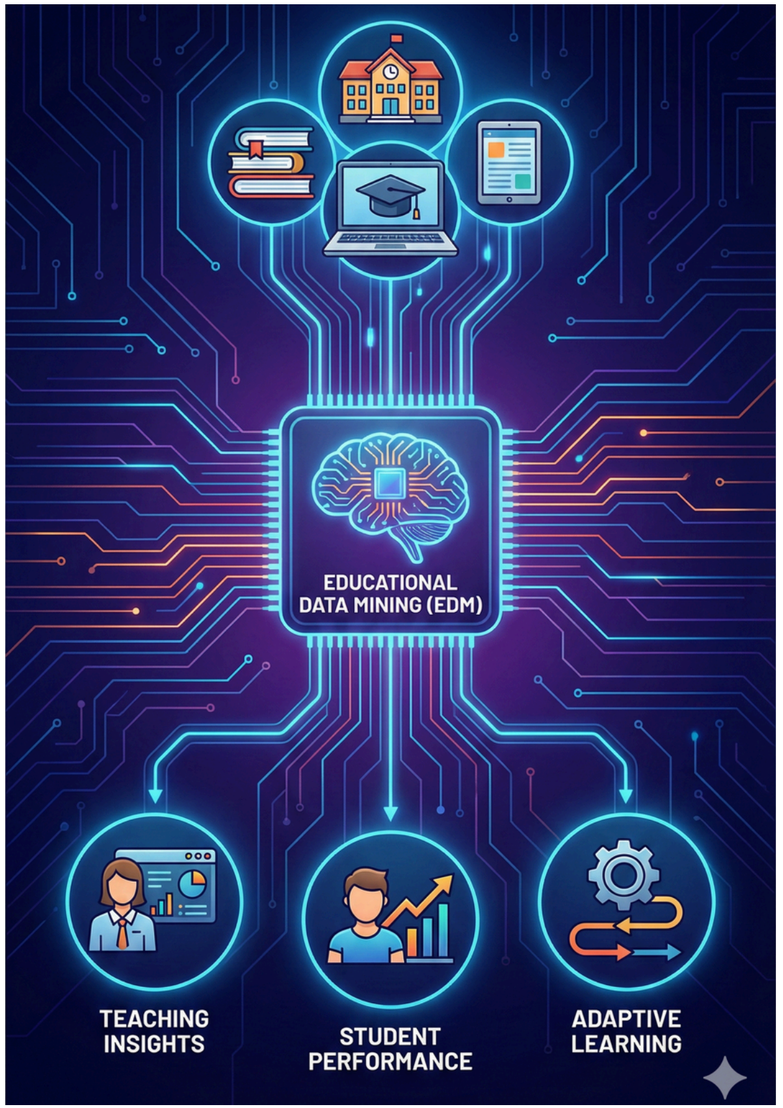
Supporting academic planning and policy decisions through data insights.

6. Assessment & Feedback Optimization

Automated essay scoring systems achieve 0.88-0.91 correlation with human graders while providing instant feedback [9].

7. Social Learning Analytics

EDM analyzes student interactions in forums and collaborative platforms to improve group learning outcomes.



CASE STUDY
MIT'S PROJECT-BASED CS LEARNING ANALYTICS SYSTEM

Problem :

- 38% dropout in advanced CS project courses [10]
- Poor team dynamics often discovered too late [11]
- Limited visibility into collaborative coding processes [10][11]

EDM Solution :

- **Git-based collaboration analytics tracking**
- **Automated project scoping advisor**
- **Real-time team health dashboard**
- **Peer feedback analysis detecting team conflict**

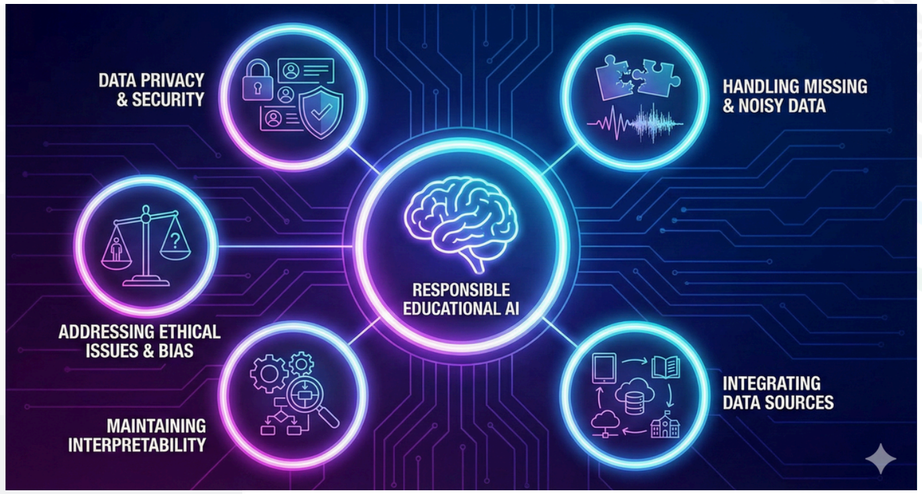
Results (2021-2023 Implementation):

- 📈 31% increase in project completion rates [10]
- 👥 42% improvement in team satisfaction scores [10]
- ⏱️ 35% reduction in instructor intervention time [11]
- 🎖️ Underrepresented student participation in leadership roles increased by 23% [10]

CHALLENGES & CONCLUSION

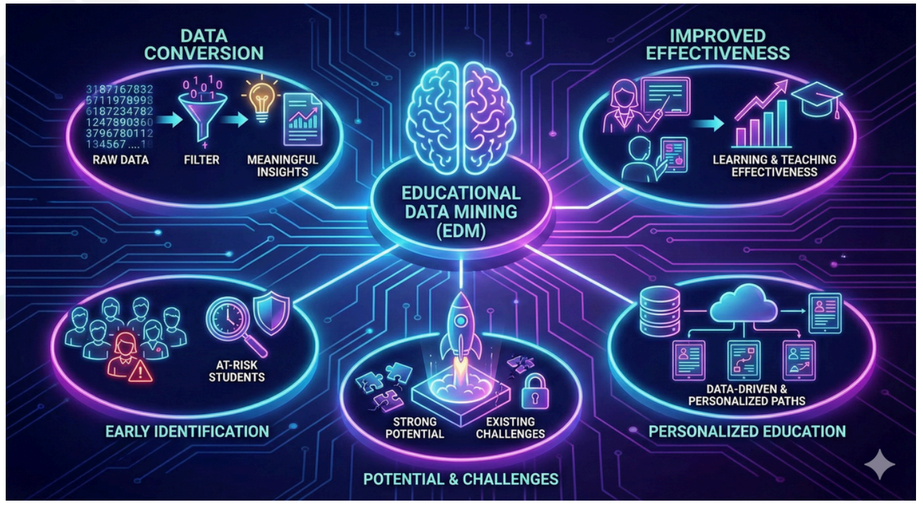
CHALLENGES

- Ensuring data privacy and security of student information
- Handling missing, noisy, and inconsistent data
- Integrating data from multiple educational sources
- Maintaining interpretability of complex models
- Addressing ethical issues and bias in predictions



CONCLUSION

- Data Mining converts raw data into meaningful insights
- Educational Data Mining improves learning and teaching effectiveness
- Enables early identification of at-risk students
- Supports data-driven and personalized education
- EDM has strong potential despite existing challenges



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Thank You